Chuyao Fu (Troy)

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Education

Southern University of Science and Technology (SUSTech) Sep. 2023 – Jun. 2027
B.Eng. in Electronic Information Engineering *GPA: 3.90/4.00 Rank: 3/46*

- **Selected Coursework:** Data Structures and Algorithm Analysis, Probability and Statistics, Machine Learning, Robotic Motion and Control, Digital Signal Processing, Signals and Systems, Analog Circuits, Digital Circuits

University of California, Berkeley Aug. 2026 – Dec. 2026
Visiting Student

Research Experience

HMI Lab, School of Computer Science, Peking University Jun. 2025 – Present
Undergraduate Researcher *Supervisor: Prof. Shanghang Zhang*

- Leading **Token-World**, which studies direct action-conditioned simulation for VLA policies in compact visual-token space without intermediate RGB reconstruction.
- Leading **VA-CLIP**, a reference-free verifier that learns action-video compatibility to evaluate and select embodied world-model rollouts.
- Contributed to **LJM**, an action-conditioned world model that predicts robot-environment interaction latents before generating future videos.
- First author on both under-review manuscripts; responsible for problem formulation, method design, experiments, and manuscript preparation.

Robotics and Computer Vision Lab, SUSTech Sep. 2024 – Present
Undergraduate Researcher *Supervisors: Prof. Hong Zhang and Prof. Max Q.-H. Meng*

- Proposed **ProDrive**, a proactive autonomous-driving framework that jointly trains a planner and a BEV world model through ego-environment co-evolution; first author and lead researcher.
- Co-led **AxisField-GS**, a feed-forward articulation-recovery framework that predicts dense Gaussian-level motion fields and recovers revolute joint axes from single-state semantic 3DGS assets; co-first author.

Selected Publications and Manuscripts

ProDrive: Proactive Planning for Autonomous Driving via Ego-Environment Co-Evolution
Chuyao Fu, Shengzhe Gan, Zhuoli Ouyang, Yuhan Rui, Xiaowei Chi, Sirui Han, Jiankun Wang, Hong Zhang
CVPR 2026 GigaBrain Challenge Workshop

Learning Action-Video Compatibility as a Reference-Free Verifier for Embodied World Models
Chuyao Fu, Xiaojie Zhang, Xiaowei Chi, Rongyu Zhang, Yuhan Rui, Jiajun Cao, Yike Guo, Sirui Han, Shanghang Zhang
Under review

Token-World: Direct World Model Simulator for Vision-Language-Action Models
Chuyao Fu, Xiaowei Chi, Haoran Li, Yuhan Rui, Zezhong Qian, Yu-Kai Wang, Xiaojie Zhang, Yunfan Lou, Hongyang Cheng, Yike Guo, Sirui Han, Shanghang Zhang
Under review

AxisField-GS: Feed-Forward Articulation Recovery from Single-State Semantic Gaussian Assets
Shengzhe Gan*, **Chuyao Fu***, Yugong Wang, Guangcheng Chen, Hong Zhang. *Equal contribution
Under review

Mask World Model: Predicting What Matters for Robust Robot Policy Learning
Yunfan Lou, Xiaowei Chi, Xiaojie Zhang, Zezhong Qian, Chengxuan Li, Rongyu Zhang, Yaoxu Lyu, Guoyu Song, **Chuyao Fu**, Haoxuan Xu, Pengwei Wang, Shanghang Zhang
ICML 2026

EchoArena: Mutual Benchmarking of World Models and VLA Policies
Yu-Kai Wang, Kevin Zhang, Xiaowei Chi, Tianxing Chen, Siqiao Huang, **Chuyao Fu**, Tiecheng Guo, Peidong Jia, Yan Qin, Kuangzhi Ge, Siyuan Qian, Weishi Mi, Zezhong Qian, Jiajun Li, Qingpo Wuwu, Shanghang Zhang, Jian Tang
CVPR 2026 GigaBrain Challenge Workshop

FORCE: Efficient VLA Reinforcement Fine-Tuning via Value-Calibrated Warm-up and Self-Distillation
Shuyi Zhang, Yunfan Lou, Hongyang Cheng, Yichen Guo, **Chuyao Fu**, Yaoxu Lyu, Xiaojie Zhang, Haoran Li, Pengwei Wang, Zhongyuan Wang, Shanghang Zhang
arXiv preprint, 2026

LJM: A Latent Joint-conditional Model for Action-Conditioned Embodied World Model
Kuangzhi Ge, Yu-Kai Wang, Yuhan Rui, Xiaojie Zhang, Hao Chen, Jiaming Liu, **Chuyao Fu**, Zhi Yang Chen, Tianze Zhou, Zhuoqun Wu, Athend Zhuoming Zhong, Chak-Wing Mak, Kevin Zhang, Xiaowei Chi, Shanghang Zhang, Sirui Han
Under review

Industry Experience

Muka Robotics May. 2026 – Present
Research Intern *Beijing*

- Developing representation learning techniques and data cleaning pipeline for foundational world-action-model pretraining.
- Built a physical world model demo in which a robot observes, predicts, and plays tabletop curling against a human.

XtalPi Mar. 2026 – Apr. 2026
Research Intern, Future Chemistry Department *Shenzhen*

- Trained and deployed VLA policies, including LingBot-VLA, π_0, and $\pi_{0.5}$, for robot manipulation in laboratory-automation settings.
- Integrated perception, policy inference, and control on real robot platforms; conducted iterative evaluation and ablations to diagnose deployment failures.

Beijing Academy of Artificial Intelligence Jun. 2025 – Sep. 2025
Research Intern, Embodied Multimodal Large Models Research Center *Beijing*

- Processed large-scale embodied datasets, including AgiBot-World and DROID, for VLA pretraining; implemented action normalization, tokenization, and structured training-data conversion.
- Reproduced representative VLA reinforcement-learning and robot-policy post-training frameworks; contributed to **FORCE** and the cross-embodiment **RoboBrain-X0** project.

Selected Projects

Awesome Representation for World Models Aug. 2026 – Present
Open-Source Literature Curation *Project Lead*

- Currently lead a collaborative survey of representations for world models, coordinating lab members to curate literature around state and transition representations.
- Designed a Cloudflare-based webpage workflow spanning literature review, taxonomy-based classification, and pull-request review for multi-contributor maintenance.

RoboBrain-X0: Cross-Embodiment Vision-Language-Action Model Jun. 2025 – Sep. 2025
Open-Source VLA Model *Contributor*

- Contributed to **RoboBrain-X0**, a unified VLA model for cross-embodiment action generation that supports heterogeneous robot platforms through a shared action-token representation.
- Focused on the data pipeline, processing AgiBot-World and DROID with action normalization, tokenization, and structured conversion for large-scale VLA pretraining.

Leadership and Service

Class Leader, Class 2303 2024 – 2025
Class Committee

- Led class operations and activities, including event planning, logistics coordination, routine administration, and award application materials; the class received honors including *Excellent Undergraduate Academic Style*, *Best Sports Team*, and *Best Popularity Class*.

House Division Lead, Street Dance Club 2024 – 2025
Student Organization

- Led the House division by coordinating weekly training with instructors and organizing two campus street dance performances, including rehearsal planning and member coordination.

Honors and Awards

- First Prize, Outstanding Student Award, SUSTech (top 5%; RMB 5,000 scholarship), Sep. 2025
- Third Prize, Outstanding Student Award, SUSTech (top 25%; RMB 1,000 scholarship), Sep. 2024
- First-Class Admission Scholarship, SUSTech (top 10%; RMB 50,000 scholarship), Sep. 2023
- Gold Medal, Chinese Chemistry Olympiad, Chinese Chemical Society (top 150 nationwide), Nov. 2022

Skills

- **Programming and ML:** Python, PyTorch, Linux, Git
- **Robotics:** ROS, real-robot VLA deployment, camera calibration, 3D vision, and point-cloud processing
- **Languages:** Chinese (native); English (TOEFL iBT 100: R25, L25, S23, W27)